

Option's Delta Entropy

1 Short Introduction

It is well-known that Shannon entropy for a random variable X is a measure of uncertainty, quantifying the average amount of "information" or "surprise" inherent in the possible outcomes of X . The Shannon entropy is defined as:

$$H(X) = - \sum_i p_i \ln(p_i)$$

where p_i represents the probability of the i -th outcome of X . Delta Δ of an option can be interpreted as a proxy for probability that the option will finish in-the-money (though often practically inaccurate).

We can try represent an entire options chain as a random variable where the deltas of individual options represent the values of this random variable. If Δ_i is the delta of the i -th option in the chain, and

$$\hat{\Delta} = \sum_{\text{options chain}} \Delta,$$

we can conceptualize the entropy of this delta portfolio as a measure of the proxy uncertainty across the various options in the chain. The entropy of the options chain can thus be written as

$$H = - \sum_i \frac{\Delta_i}{\hat{\Delta}} \ln \left(\frac{\Delta_i}{\hat{\Delta}} \right)$$

Here we use normalized deltas so that the sum of all normalized deltas in an options chain sum to one, allowing for a Shannon entropy interpretation. That is, the normalized delta of the i -th option is represented by $\frac{\Delta_i}{\hat{\Delta}}$. This could be used to proxy uncertainty in markets regarding expiration of certain options, and consequently H can be used as a proxy to larger price moves due to the markets needing to re-hedge more (perhaps).

It may be interesting to find the rate of change of entropy with respect to S , so we differentiate H as follows,

$$\frac{dH}{dS} = - \sum_i \left(\frac{d}{dS} \left(\frac{\Delta_i}{\hat{\Delta}} \right) \ln \left(\frac{\Delta_i}{\hat{\Delta}} \right) + \frac{\Delta_i}{\hat{\Delta}} \frac{d}{dS} \ln \left(\frac{\Delta_i}{\hat{\Delta}} \right) \right)$$

Using the chain rule, the derivative of $\ln \left(\frac{\Delta_i}{\hat{\Delta}} \right)$ with respect to S is

$$\frac{d}{dS} \left[\ln \left(\frac{\Delta_i}{\hat{\Delta}} \right) \right] = \frac{\hat{\Delta}}{\Delta_i} \frac{d}{dS} \left(\frac{\Delta_i}{\hat{\Delta}} \right)$$

Substituting this into the expression for $\frac{dH}{dS}$, we get,

$$\frac{dH}{dS} = - \sum_i \left(\frac{d}{dS} \left(\frac{\Delta_i}{\hat{\Delta}} \right) \ln \left(\frac{\Delta_i}{\hat{\Delta}} \right) + \frac{\Delta_i}{\hat{\Delta}} \frac{\hat{\Delta}}{\Delta_i} \frac{d}{dS} \left(\frac{\Delta_i}{\hat{\Delta}} \right) \right)$$

Simplifying,

$$\frac{dH}{dS} = - \sum_i \frac{d}{dS} \left(\frac{\Delta_i}{\hat{\Delta}} \right) \left(\ln \left(\frac{\Delta_i}{\hat{\Delta}} \right) + 1 \right)$$

Since the normalized deltas sum to one,

$$\sum_i \frac{\Delta_i}{\hat{\Delta}} = 1$$

we also have

$$\sum_i \frac{d}{dS} \left(\frac{\Delta_i}{\hat{\Delta}} \right) = 0$$

Therefore, the +1 term vanishes in the summation, so we can write,

$$\frac{dH}{dS} = - \sum_i \frac{d}{dS} \left(\frac{\Delta_i}{\hat{\Delta}} \right) \ln \left(\frac{\Delta_i}{\hat{\Delta}} \right)$$

Now, since $\hat{\Delta}$ is itself a function of S , the normalisation factor must also be differentiated. Using the quotient rule,

$$\frac{d}{dS} \left(\frac{\Delta_i}{\hat{\Delta}} \right) = \frac{\frac{d\Delta_i}{dS} \hat{\Delta} - \Delta_i \frac{d\hat{\Delta}}{dS}}{\hat{\Delta}^2}$$

Substituting this into the expression for $\frac{dH}{dS}$, we get

$$\frac{dH}{dS} = - \sum_i \left[\frac{\frac{d\Delta_i}{dS} \hat{\Delta} - \Delta_i \frac{d\hat{\Delta}}{dS}}{\hat{\Delta}^2} \right] \ln \left(\frac{\Delta_i}{\hat{\Delta}} \right)$$

Equivalently, since $\frac{d\Delta_i}{dS}$ is the gamma of the i -th option, we can write

$$\frac{dH}{dS} = - \sum_i \left[\frac{\Gamma_i \hat{\Delta} - \Delta_i \hat{\Gamma}}{\hat{\Delta}^2} \right] \ln \left(\frac{\Delta_i}{\hat{\Delta}} \right)$$

The sum runs over all options in the chain, so the total sensitivity of entropy to changes in S depends on how the normalized deltas of all options evolve as S changes. For a single option, this simplifies to,

$$H = -\frac{\Delta}{\hat{\Delta}} \ln \left(\frac{\Delta}{\hat{\Delta}} \right) = -1 \ln(1) = 0$$

and therefore

$$\frac{dH}{dS} = 0.$$

It should also be noted that, for a strict Shannon entropy interpretation, the normalized deltas $\frac{\Delta_i}{\hat{\Delta}}$ must be non-negative and sum to one. This requires consideration if we include signed deltas, such as put deltas.